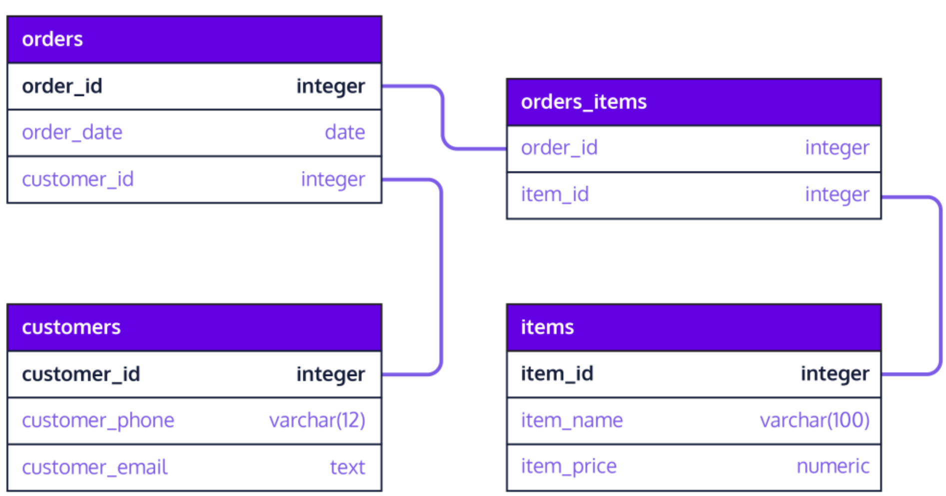
Introduction To Database System
Unit – 2

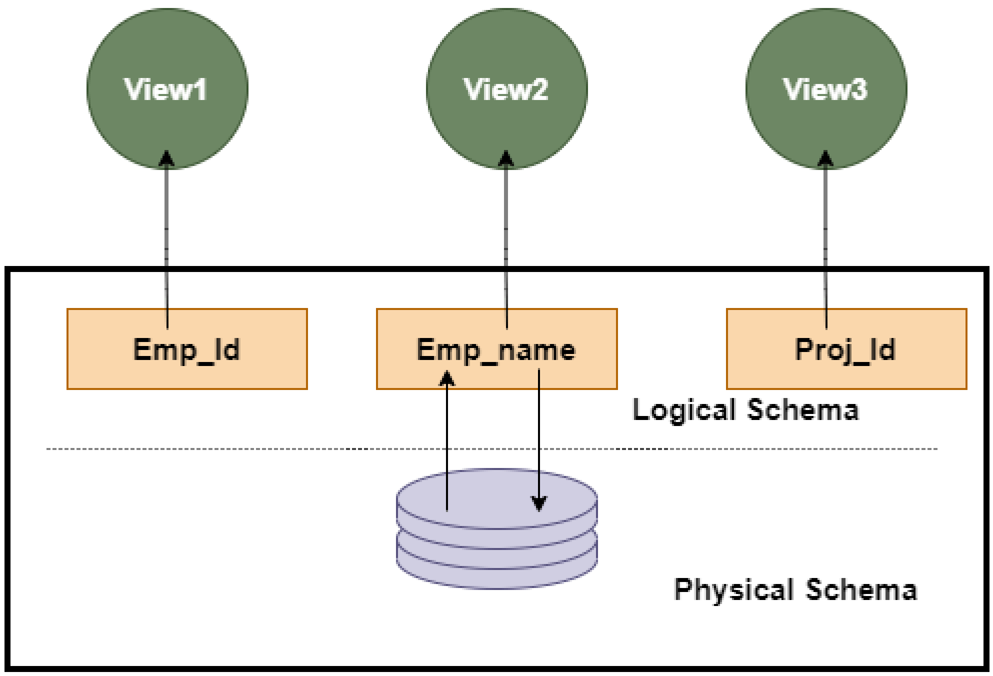
Schema - The overall design of a database is called schema which is specified during database design and changes rarely.

- A database schema is the skeleton structure of the database.
- A database schema is designed by database designers to help programmers whose software will interact with the database.
- A database schema contains schema objects that may include tables, fields, packages, views, relationships, primary key, foreign key.



Types of Database Schema

1. Physical Schema
2. Logical Schema
3. View Schema



1. Physical Database Schema/Internal Schema

It specifies how the data is stored physically on a storage system or disk storage in the form of Files and Indexex. Designing a database at the physical level is called a **physical schema**.

2Logical Database Schema/Conceptual Schema

It specifies all the logical constraints that need to be applied to the stored data. It defines the views, integrity constraints, and table. The logical schema represents how the data is stored in the form of tables and how the attributes of a table are linked together.

3. View Schema/External Schema

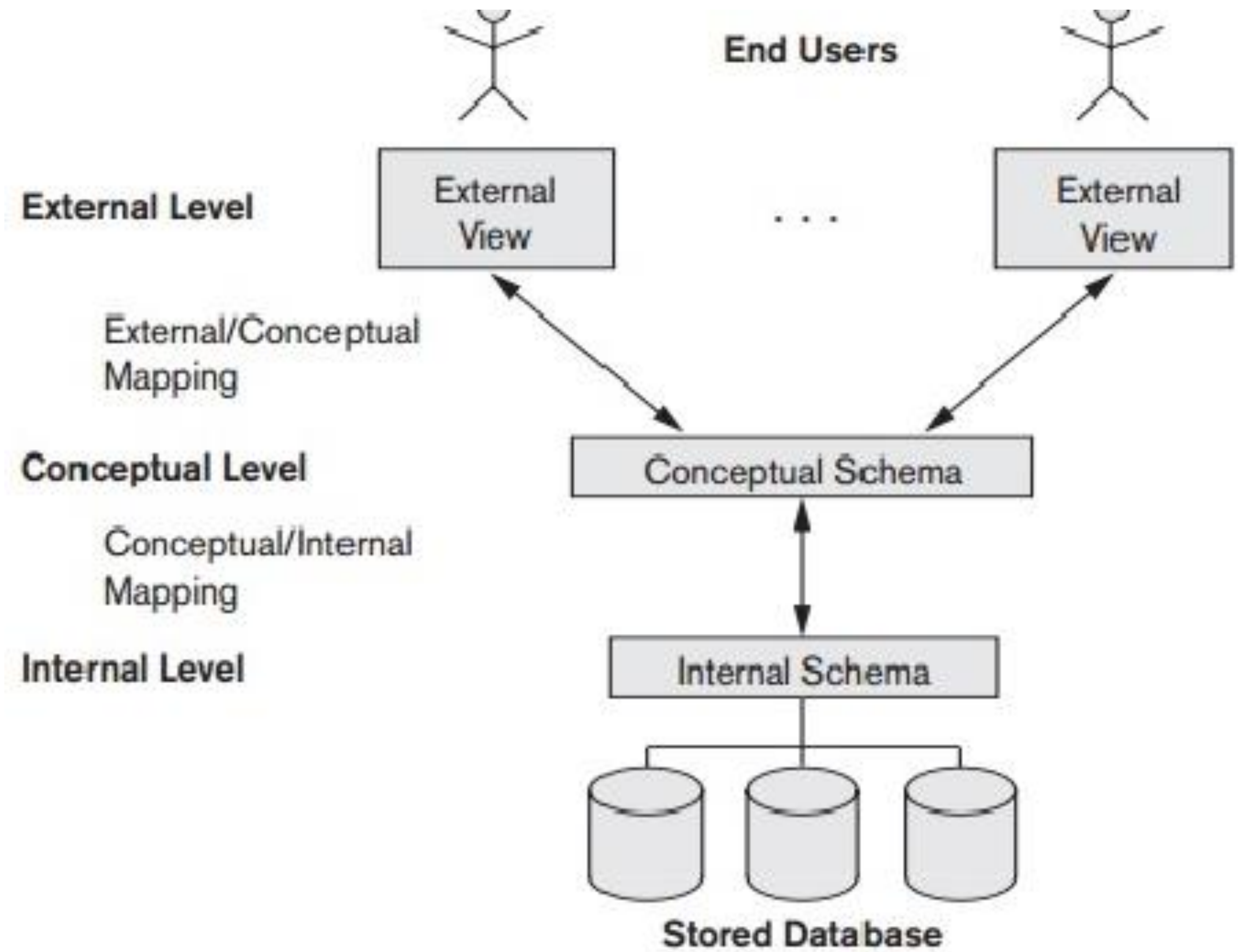
The view level design of a database is known as **view schema**. This schema generally describes the end-user interaction with the database systems.

Instance

Data in the database is of dynamic nature. It keeps changing from time to time. The collection of data stored in a database at a particular moment of time is called the instance of a database. It is also called database state or snapshot.

Three Schema Architecture

The three-schema architecture divides the database into three-level used to create a separation between the physical database and the user application. In simple terms, this architecture hides the details of physical storage from the user.

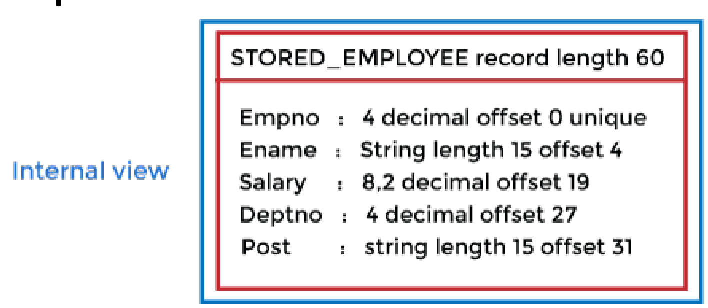


1. Internal Level

- The internal level has an internal schema which describes the physical storage structure of the database.
- The internal schema is also known as a physical schema.
- It uses physical data model. It is used to define that how the data will be stored in a block.

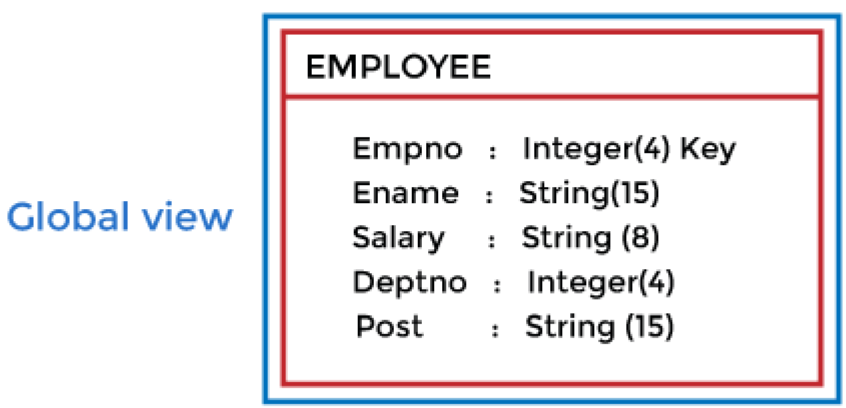
The internal level is generally is concerned with the following activities:

- Storage space allocations.
  For Example: B-Trees, Hashing etc.
- Access paths.
  For Example: Specification of indexes, pointers and sequencing.
- Data Structure
- Data compression and encryption techniques.
- Optimization of internal structures.
- Representation of stored fields.

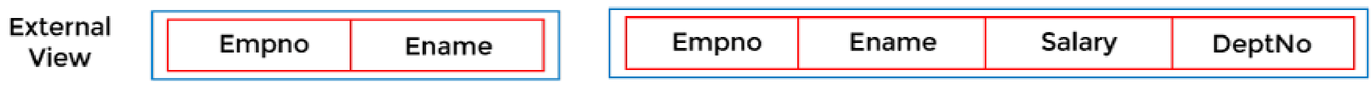


2. Conceptual Level

- The conceptual schema describes the design of a database at the conceptual level.
- Conceptual level is also known as logical level.
- The conceptual level describes what data are to be stored in the database and also describes what relationship exists among those data.
- It describes entities, data types, relationships and constraints on the data.
- In this level, internal details such as an implementation of data structure are hidden.
- Programmers and database administrators work at this level.



3. External Level



- At the external level, a database contains several schemas that sometimes called as subschema. The subschema is used to describe the different view of the database.
- An external schema is also known as view schema.
- Each view schema describes the database part that a particular user group is interested and hides the remaining database from that user group.
- Individual users are given different views according to their requirements.
- The view schema describes the end user interaction with database systems.
- View level increases the security of database.
- It provides many views of the same database.

Objectives of Three Schema Architecture

The main objective of three level architecture is to enable multiple users to access the same data with a personalized view while storing the underlying data only once. Thus it separates the user's view from the physical structure of the database. This separation is desirable for the following reasons:

- Different users need different views of the same data.
- The approach in which a particular user needs to see the data may change over time.
- The users of the database should not worry about the physical implementation and internal workings of the database such as data compression and encryption techniques, hashing, optimization of the internal structures
- All users should be able to access the same data according to their requirements.
- DBA should be able to change conceptual structure of database without affecting user's
- Internal structure of the database should be unaffected by changes to logical aspects of the database.

Data Independence

A database system normally contains a lot of data in addition to users' data. For example, it stores data about data, known as metadata, to locate and retrieve data easily. It is rather difficult to modify or update a set of metadata once it is stored in the database. But as a DBMS expands, it needs to change over time to satisfy the requirements of the users. If the entire data is dependent, it would become a tedious and highly complex job.

Data independence refers to characteristic of being able to modify the schema at one level of the database system without changing the schema at the next higher level.

There are two types of data independence:

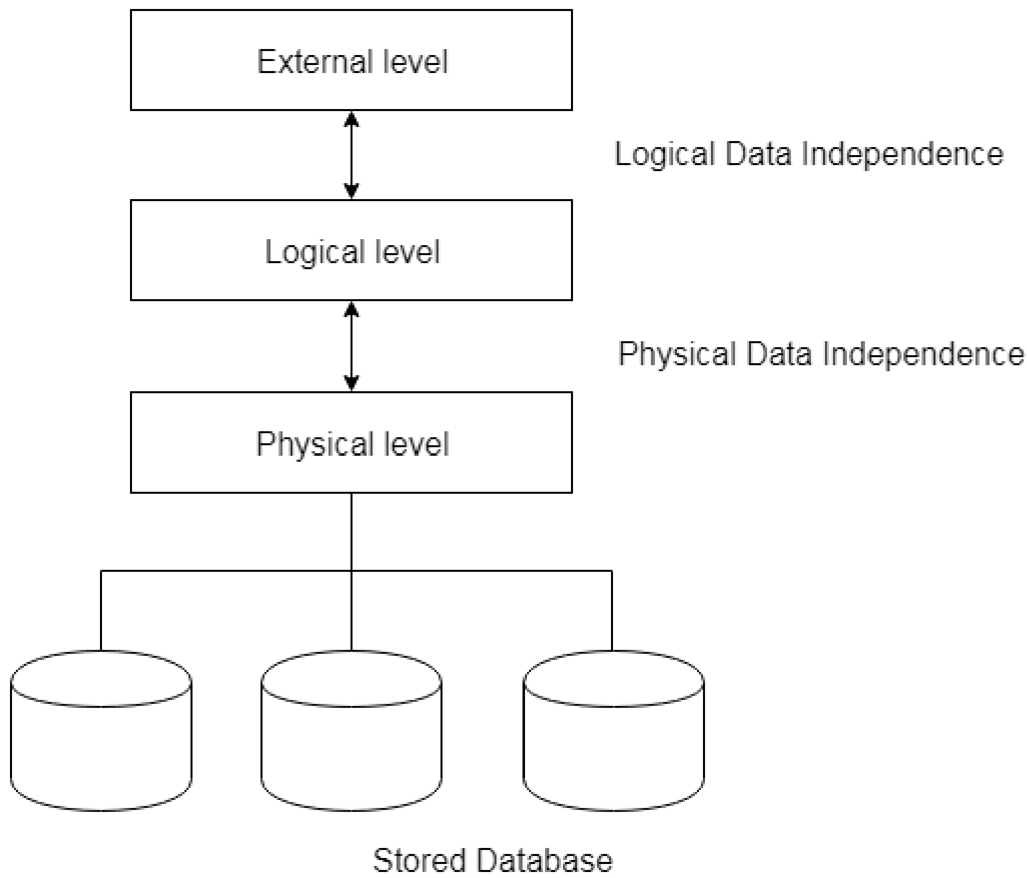
1. Physical Data Independence

- It can be defined as the capacity to change the internal schema without having to change the conceptual schema.
- If we do any changes in the storage size of the database system server, then the Conceptual structure of the database will not be affected.

- Physical data independence is used to separate conceptual levels from the internal levels.
  For example in case we want to change or upgrade storage system itself − suppose we want to replace hard-disks with SSD − it should not have any impact on logical data or schemas.

2. Logical Data Independence

- It refers characteristic of being able to change the conceptual schema without having to change the external schema.
- Logical data independence is used to separate the external level from the conceptual view.
- If we do any changes in conceptual view of data then user view of data would not be affected.
- Example: If there is a database of banking system and we want to add details of a new customer or we want to update data of a customer, at logical level data will be changed but it will not affect the Physical level or structure of the database.



Mapping

The three levels of DBMS architecture don't exist independently of each other. There must be correspondence between the three levels i.e. how they actually correspond with each other. DBMS is responsible for correspondence between the three types of schema. This correspondence is called **Mapping.**

There are basically two types of mapping in the database architecture:

1.Conceptual/ Internal Mapping

The Conceptual/ Internal Mapping lies between the conceptual level and the internal level. Its role is to define the correspondence between the records and fields of the conceptual level and files and data structures of the internal level.

2.External/ Conceptual Mapping

The External/Conceptual Mapping lies between the external level and the Conceptual level. Its role is to define the correspondence between a particular external and the conceptual view.

<u>Classification of Database Management System</u>

The DBMS can be classified into different categories on the basis of several criteria such as the data model they are using, number of users they support, number of sites over which database is distributed and the purpose they serve.

Based on Data Models

The data model defines the physical and logical structure of a database which involves the data types, the relationship among the data, constraints applied on the data and even the basic operations specifying retrieval and updation of data in the database. Depending upon how the data is structured, data models are further classified into:

1.Record Based Logical Models

a. Hierarchical Model

b. Network Model

c. Relational Model

2.Object based Data Models

a. E-R Model

b. Object oriented Model

Based on Number of Users

DBMS can either be used by a single user or it can be used by multiple users.

- **single-user system :** The database system that can be used by a single user at a time is referred to **as a single-user system**

- **multiple user system:** The database system that can be used by multiple users at a time is referred to as a **multiple user system**.

Based on Database Distribution

Depending on the distribution of the database over numerous sites we can classify the database as:

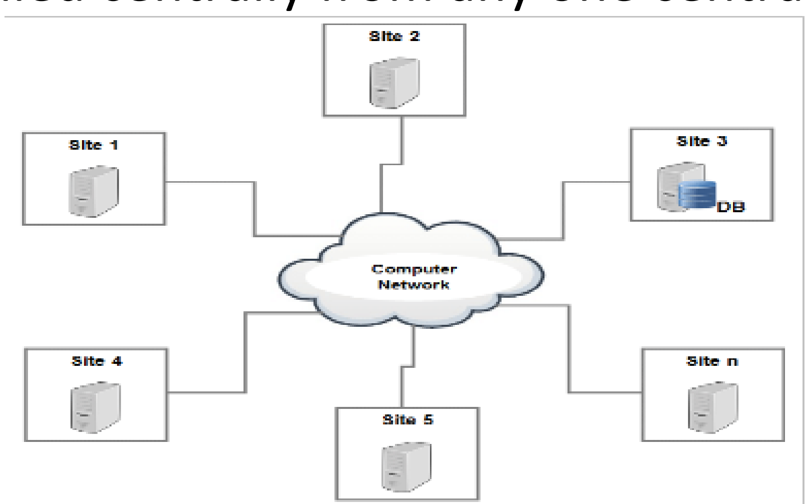
- **Centralized DBMS**
  In the centralized DBMS, the entire database is stored in a single computer site. Though the centralized database support multiple users still the DBMS software and the data both are stores on a single computer site.

- **Distributed DBMS**
  In the distributed DBMS (DDBMS) the database and the DBMS software are distributed over many computer sites. These computer sites are connected via a computer network.
- The DDBMS is further classified as homogeneous DDBMS and heterogeneous DDBMS.

- **Homogeneous DDBMS:** The homogeneous DDBMS has the same DBMS software at all the distributed sites.
- **Heterogeneous DDBMS:** The heterogeneous DDBMS has different DBMS software for different sites.

Centralized DBMS Architecture

- The centralized database system consist of a single server with associated data storage devices and other peripherals.
- It is physically confined to a single location.
- The data can be accessed from the multiple sites with the use of a computer network while the database is maintained at the central site.
- The management of system and its data are controlled centrally from any one central site.



Advantages

1. Single site provides high degree of security, concurrency, backup and recovery control.
2. Centralized control of data avoids unnecessary duplication of data.
3. Reduces total amount of data storage required.
4. More cost effective than other types of database systems as labour, power supply and maintenance costs are all minimized.

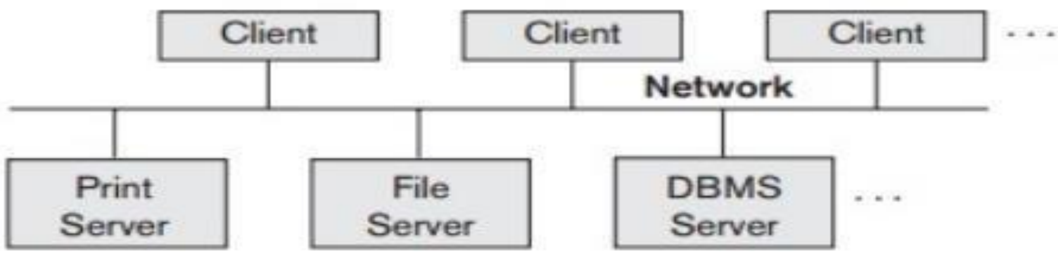
Disadvantages

1. When the central site computer or database system goes down, then everyone is blocked from using the system until the system comes back.
2. Communication costs from the terminals to the central site can be expensive.
3. Since there is minimal data redundancy, if a set of data is unexpectedly lost it is very hard to retrieve it back.

Client/Server Architectures

- The client/server architecture was developed to deal with computing environments in which a large number of PCs, workstations, file servers, printers, database servers.
- Web servers, e-mail servers, and other software and equipment are connected via a network. The idea is to define specialized servers with specific functionalities.
- For example, it is possible to connect a number of PCs or small workstations as clients to a file server that maintains files of client machines.
- Another machine can be designated as a printer server by being connected to various printers; all print requests by the clients are forwarded to this machine.
- Web servers or e-mail servers also fall into the specialized server category. The resources provided by specialized servers can be accessed by many client machines.
- The client machines provide the user with the appropriate interfaces to utilize these servers, as well as with local processing power to run local applications.

A. **Two-Tier Client Server Architecture**



- The concept of client/server architecture assumes an underlying framework that consists of many PCs and workstations as well as a smaller number of mainframe machines, connected via LANs and other types of computer networks.
- A client in this framework is typically a user machine that provides user interface capabilities and local processing.
- When a client requires access to additional functionality— such as database access—that does not exist at that machine, it connects to a server that provides the needed functionality.
- A server is a system containing both hard-ware and software that can provide services to the client machines, such as file access, printing, archiving, or database access.

- The user interface programs and application programs can run on the client side. When DBMS access is required, the program establishes a connection to the DBMS (which is on the server side)
- Once connection is created, the client program can communicate with the DBMS. A standard called **Open Database Connectivity** (**ODBC**) provides an **application programming interface** (**API**), which allows client-side programs to call DBMS, as long as both client and server machines have necessary software installed.
- Any query results are sent back to the client program, which can process and display the results as needed.

Advantages

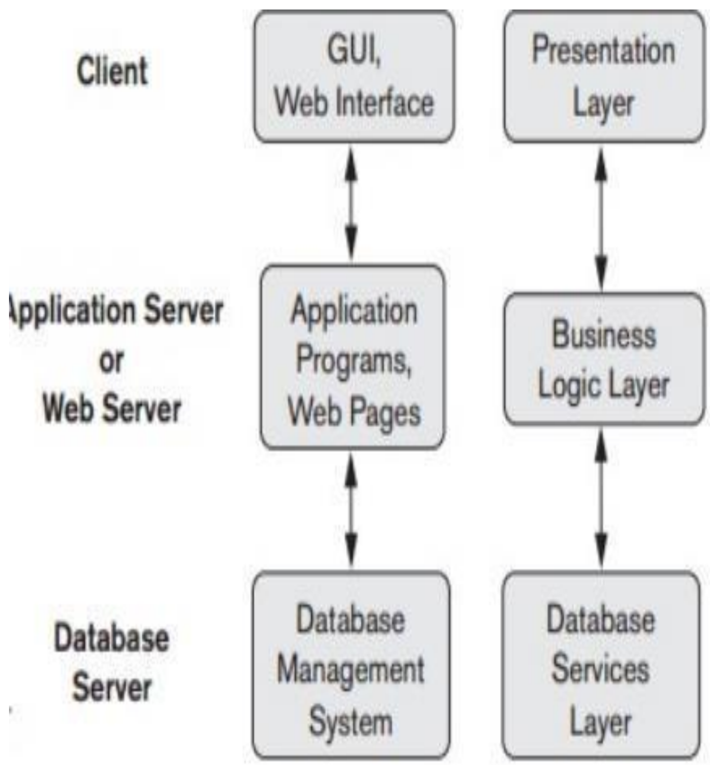
- This architecture provides simplicity and seamless compatibility with existing systems.
- It performs well with moderate number of clients.

Disadvantage

- Poor performance when large no of clients submit their request.

B. <u>Three-tier Client Server Architecture</u>

- Many Web applications use an architecture called the **three-tier architecture**, which adds an intermediate layer between the client and the database server.
- This intermediate layer or **middle tier** is called the **application server** or the **Web server**, depending on the application.
- The client communicates with application server, which in turn communicates with database server.
- This server plays an intermediary role by running application programs and storing business rules (procedures or constraints) that are used to access data from the database server.
- When a client request for information , the application server accepts the request, process it and sends corresponding database commands to database server. The database server sends the result back to application server which is converted into GUI format and presented to the client. **Thus, the user interface, application rules, and data access act as the three tiers.**



Advantages

- Advances in encryption and decryption technology make it safer to transfer sensitive data from server to client in encrypted form, where it will be decrypted.
- This technology gives provides increased performance.
- It provides flexibility, maintainability, scalability.
- It can also improve database security by checking a client's credentials before forwarding a request to the data-base server.

Data models

- Data models define how the logical structure of a database is modeled.
- Data Models are fundamental entities to introduce abstraction in a DBMS.
- Data models define how data is connected to each other and how they are processed and stored inside the system
- The purpose of a data model is to represent data and to make the data understandable.
- There have been many data models. They fall into three broad categories:
    - **Record Based Data Models**
    - **Physical Data Models**
    - **Object Based Data Models**

1. **Record Based Logical Models**
    - Record based models are so named because the database is structured in fixed format records of several types. Each record type defines a fixed number of fields, or attributes, and each field is usually of a fixed length.

The three most widely accepted record based data models are:

- Relational Model
- Hierarchical Model
- Network Model

a)Relational Data Model

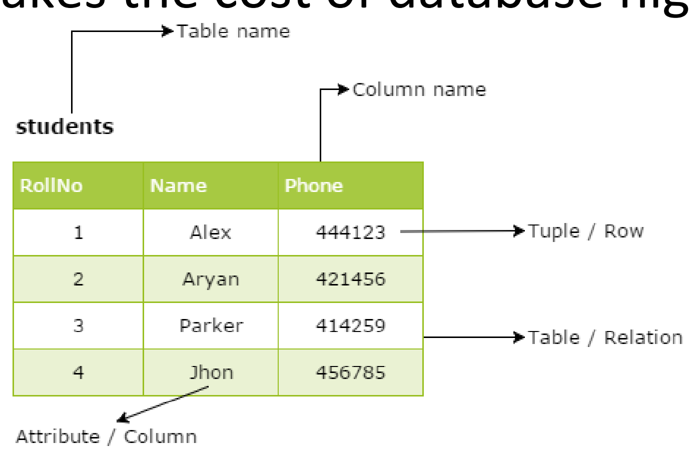
- In the relational data model, we use tables to represent data and the relationship among that data.
- Each of the tables in the relational data model has a unique name.
- A table has multiple columns where each column name is unique. A table holds records which has value for each column of the table.
- We also refer to the relational database model as a record-based data model as it holds -records of fixed-format.
- The relational database model is the most currently used data model.

Advantages

1. Relational Model are used to manage large amounts of data, such as customer information, employee data, and financial records.
2. It allows users to quickly and easily access data from multiple tables.
3. Data is organized in a structured way, it is easier to update and maintain.
4. Relational Models are most commonly used DBMS and used in wide variety of applications from banking systems to retail websites.
5. Examples of relational DBMS are Oracle, MySQL.

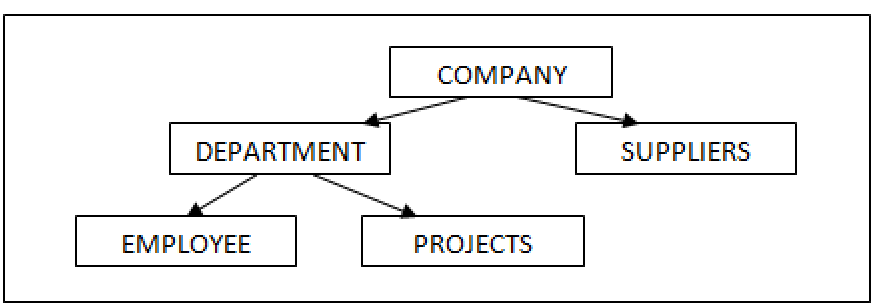
Disadvantages

1. High hardware cost: - In order to separate the physical data information from the logical data, more powerful system hardware – memory is required. This makes the cost of database high.



b)Hierarchical Model

- A hierarchical database model is a data model in which the data is organized into a tree-like structure.
- The data is stored as records which are connected to one another through links.
- A record is a collection of fields, with each field containing only one value.
- The entity type of a record defines which fields the record contains.
- A record in the hierarchical database model corresponds to a row (or tuple) in the relational database model and an entity type corresponds to a table (or relation).
- In order to retrieve data from a hierarchical database the whole tree needs to be traversed starting from the root node.



Advantages

- It helps to address the issues of flat file data storage.
- In flat files, data will be scattered and there will not be any proper structuring of the data.
- This model groups the related data into tables and defines the relationship between the tables, which is not addressed in flat files.

Disadvantage

- It fails to handle many to many relationships efficiently. It results in redundancy and confusion. It can handle only parent-child kind of relationship.
- If we need to fetch any data in this model, we have to start from the root of the model and traverse through its child till we get the result.
- In order to perform the traversing, either we should know well in advance the layout of model or we should be very good programmer. Hence fetching through this model becomes bit difficult.

c)The Network model

- This is the enhanced version of hierarchical data model. It is designed to address the drawbacks of the hierarchical model.
- This data model is also represented as hierarchical, but this model will not have single parent concept.
- Any child in the tree can have multiple parents
- The Network model replaces the hierarchical tree with a graph thus allowing more general connections among the nodes.
- The main difference of the network model from the hierarchical model, is its ability to handle many to many (N: N) relations.
- Allows a record to have more than one parent.

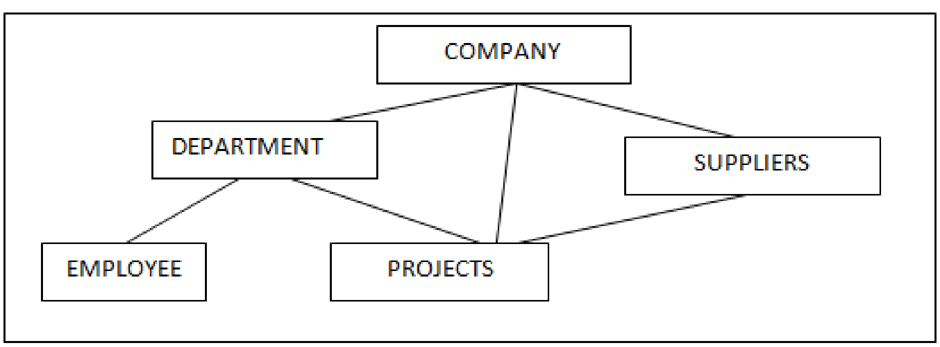
Advantages

- This model has ability to manage one-to-one (1:1) and many-to-many (N: N) relationships.
- Accessing the data is simpler when compared to the hierarchical model.
- Data Integrity: In a network model, there's always a connection between the parent and

the child segments because it depends on the parent-child relationship

Disadvantages

- There is no independence between any objects. Hence any changes to object will need change whole model. Hence difficult to manage.
- It would be little difficult to design the relationship between the entities, since all the entities are related in some way.
- System Complexity: Each and every record has to be maintained with the help of pointers, which makes database structure more complex.
- Functional Flaws: Because a great number of pointers is essential, insertion, updates, and deletion become more complex.



2. Object Based Data Models

- In object based data models, the focus is on how data is represented. The data is divided into multiple entities each of which have some defining characteristics.
- Moreover, these data entities are connected with each other through some relationships.
- So, in object based data models the entities are based on real world models, and how the data is in real life.

a) **Entity Relationship Data Model**

ER model is used to represent real life scenarios as entities. The properties of these entities are their attributes in the ER diagram and their connections are shown in the form of relationships.

b) **Object Oriented Data Model**

Object oriented data model is also based on using real life scenarios. In this model, the scenarios are represented as objects. The objects with similar functionalities are grouped together and linked to different other objects.

Advantages

- Because of its inheritance property, we can re-use the attributes and functionalities. It reduces the cost of maintaining the same data multiple times. Hence it reduces the overhead and maintenance costs.
- It becomes more flexible in the case of any changes.
- Codes are re-used because of inheritance

Disadvantages

- It is not widely developed and complete to use it in the database systems. Hence it is not accepted by the users.
- It is an approach for solving the requirement. It is not a technology. Hence it fails to put it in the database management systems.

Conceptual Data Model

- Conceptual data model, describes the database at a very high level and is useful to understand the needs or requirements of the database.
- It is this model, that is used in the requirement gathering process i.e., before the Database

Designers start making a particular database. One such popular model is the (ER model).

- The ER model specializes in entities, relationships and even attributes which are used by the database designers.

Physical Data Model

- Ultimately, all data in a database is stored physically on a secondary storage device such as discs and tapes.
- This is stored in the form of files, records and certain other data structures.
- It has all the information of the format in which the files are present and the structure of the databases, presence of external data structures and their relation to each other.